What is effective power?

Power adjusted for the average benefit of critical hits.

Start with two possible outcomes

A normal hit uses your Power with a damage multiplier of 1. A critical hit uses the same Power multiplied by your total critical-damage multiplier.

Let **P** be Power, **c** be critical chance as a fraction, and **m** be the total critical-damage multiplier. For example, 75% crit chance means c = 0.75, and 230% critical damage means m = 2.30.

A normal hit occurs with probability **1 − c** and has power-equivalent value P. A critical hit occurs with probability **c** and has value P × m.

Take the probability-weighted average

Multiply each outcome by how often it happens, then add them:

$$E = (1 - c)P + cPm$$
$$E = P \times [(1 - c) + cm]$$
$$E = P \times [1 + c(m - 1)]$$

Why subtract 1 from critical damage?

The leading 1 already counts normal damage on every hit. A critical hit at 230% adds another **130%** on top of that normal damage, so its bonus multiplier is 2.30 − 1 = 1.30. Adding the full 2.30 again would count the normal portion twice.

Example: 3000 Power, 75% crit, 230% crit damage

Imagine 100 otherwise identical hits, with exactly 75 critical hits and 25 normal hits:

$$\text{Normal contribution: } 25 \times 3000 = 75{,}000$$
$$\text{Critical contribution: } 75 \times (3000 \times 2.30) = 517{,}500$$
$$\text{Average: } (75{,}000 + 517{,}500) \,/\, 100 = 5925$$

The formula gives the same result directly:

$$E = 3000 \times [1 + 0.75(2.30 - 1)] = 5925$$

This is the long-run expected average, not a guarantee that every set of 100 hits contains exactly 75 crits.

What the number does and does not mean

At 0% crit chance, E = P. At 100%, E = P × m. Effective power is a comparison metric, not a character-sheet stat or actual damage per second. Weapon strength, skill coefficients, enemy armor, attack frequency and other modifiers are not included.

Balancing Power and Ferocity

A practical rule, its mathematical basis, and worked examples.

The rule of thumb

Compare equal amounts of **Power versus Ferocity**, using your buffed, in-combat stats. Let **D** be displayed critical damage in percent: for example, D = 230 means 230%.

> **At 100% crit chance: Power target = 15 × D**
> **At 75% crit chance: Power target = 15 × D + 500**

- **Below the target:** adding Power gives more average strike damage.
- **Above the target:** adding Ferocity gives more average strike damage.
- **At the target:** equal-sized additions give the same benefit.

For 230% critical damage, the target is **3450 Power at 100% crit chance**, or **3950 Power at 75%**.

Why the balance exists

Power improves every hit. Ferocity improves critical hits only, but its benefit grows with your existing Power. Conversely, higher critical damage makes each added point of Power more valuable.

Let **P** be Power and **c** be critical chance as a fraction (75% = 0.75). Average effective power is:

$$\mathrm{E} = \mathrm{P} \times [(1 - \mathrm{c}) + \mathrm{c} \times \mathrm{D} \,/\, 100]$$

The first term accounts for normal hits; the second accounts for critical hits. The displayed critical damage is the **total** multiplier: 230% means 2.30×, not 3.30×.

Deriving the general rule

At 15 Ferocity per percentage point of critical damage, one Ferocity adds 1/1500 to the critical-hit multiplier. Therefore:

$$\text{Gain per Power} = (1 - \mathrm{c}) + \mathrm{cD} \,/\, 100$$
$$\text{Gain per Ferocity} = \mathrm{Pc} \,/\, 1500$$

Set the two gains equal and solve for Power:

$$\mathrm{P} = 15\mathrm{D} + 1500(1 - \mathrm{c}) \,/\, \mathrm{c}$$

At c = 1, the extra term is zero. At c = 0.75, it is 1500 × 0.25 / 0.75 = **500**. This extra Power reflects its advantage on the 25% of hits that do not crit.

WORKED EXAMPLES

Putting the rule to work

75% crit chance, 230% crit damage

Out of many hits, 25% deal normal damage and 75% deal 2.3× damage. The average multiplier is:

$$0.25 \times 1 + 0.75 \times 2.3 = 1.975$$

Option A: add 250 Power

$$\text{Effective-power gain} = 250 \times 1.975 = 493.75$$

This gain is the same regardless of starting Power, as long as the critical stats stay fixed.

Option B: add 250 Ferocity

250 Ferocity adds 250/15 = 16.666... percentage points of critical damage, increasing the critical-hit multiplier by exactly 1/6.

$$\text{Effective-power gain} = \text{P} \times 0.75 \times (1/6) = 0.125\text{P}$$

Setting 493.75 = 0.125P gives the break-even point: **P = 3950**.

Starting Power	Gain: +250 Power	Gain: +250 Ferocity	Better choice
3000	493.75	375.00	Power
3950	493.75	493.75	Equal
4500	493.75	562.50	Ferocity

Your example: 100% crit, 231% crit damage

The balance point is 15 × 231 = **3465 Power**. Below are the final effective-power totals for each upgrade:

Base Power	+250 Power	+250 Ferocity	Power advantage
2700	6814.50	6687.00	1.9067%
3450	8547.00	8544.50	0.0293%

Percentage advantage = (Power-option total / Ferocity-option total − 1) × 100. Calculations use unrounded values; only displayed results are rounded.

Use the rule correctly

- Use **in-combat Power, crit chance and crit damage**, including relevant buffs. Effective crit chance cannot exceed 100%.
- Keep crit chance fixed when comparing these two upgrades. If gear changes several stats, calculate each complete setup instead.
- This compares equal additions, such as +250 Power versus +250 Ferocity. Moving existing stats between them requires recalculating the balance as both change.
- Effective power is an average strike-damage comparison, not DPS. Conditions, on-crit effects, special damage rules and rotation changes are not included.